Month-End Downtime Constraint

Kalahari Biodiversity Research Centre | Project ID: CMPG325-2026-107 | Client ID: CLI-107

Constraint

No network downtime is tolerated during month-end processing.

How the design reduces the risk

- **VLAN separation:** Admin, Research, Servers and Voice each have their own VLAN and subnet. A problem in one department, such as a faulty PC or a misbehaving phone, stays inside that VLAN and does not affect Admin traffic.
- **Dedicated server VLAN:** DNS and DHCP run on a separate Servers VLAN. Staff can still reach resources by name and get addresses automatically.
- **Voice on its own VLAN:** Adding VoIP phones (CR11) does not add load to the Admin data network.
- **Simple, tested configuration:** Every VLAN, trunk, DHCP pool and DNS record was tested before sign-off, which lowers the chance of faults during month-end.

Limitations and recommended improvements

The current design has single points of failure. Router0 and Switch0 are single devices, so if either failed, the network would go down. To fully meet a zero-downtime requirement, the recommended improvements are:

1. A second router with a redundancy protocol such as HSRP.
2. A second core switch.
3. Redundant links between switches, using spanning tree to prevent loops.

These were not built, to keep the project within its scope and the brief's wording. They are recommended for a production deployment.